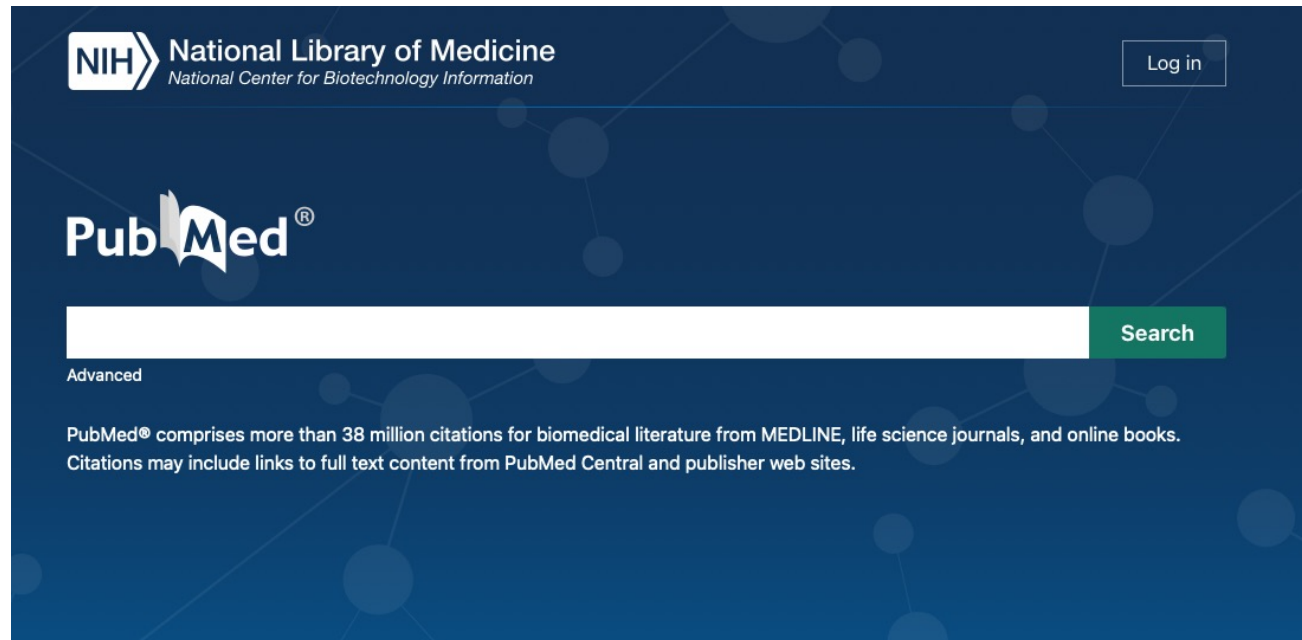


Public DATABASEs as data resources

PubMed

<https://pubmed.ncbi.nlm.nih.gov>

PubMed



<https://pubmed.ncbi.nlm.nih.gov>

The Cancer Genome Atlas Program (TCGA)

 **NATIONAL CANCER INSTITUTE**
GDC Data Portal

[Video Guides](#) [Send Feedback](#) [Browse Annotations](#) [Manage Sets](#) [Cart](#) [Login](#) [GDC Apps](#)

[Analysis Center](#) [Projects](#) [Cohort Builder](#) [Repository](#)

Genomic Data Commons Data Portal

Harmonized Cancer Datasets
A repository and computational platform for cancer researchers who need to understand cancer, its clinical progression, and response to therapy.

[Explore Our Cancer Datasets](#)

Data Portal Summary

[Data Release 42.0 - January 30, 2025](#)

**86
Projects**

**69
Primary Sites**

**44,736
Cases**

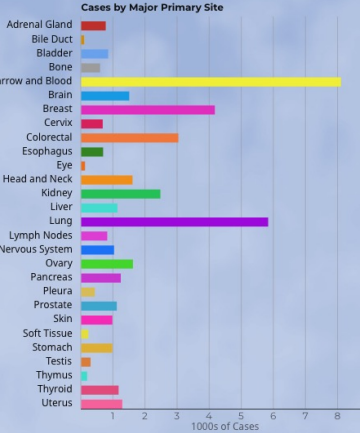
**1,121,816
Files**

**22,534
Genes**

**2,940,240
Mutations**



Cases by Major Primary Site



Major Primary Site	1000s of Cases
Adrenal Gland	0.5
Bile Duct	0.2
Bladder	0.5
Bone	0.5
Bone Marrow and Blood	7.5
Brain	1.5
Breast	4.0
Cervix	0.5
Colorectal	3.0
Esophagus	0.5
Eye	0.2
Head and Neck	1.5
Kidney	2.5
Liver	1.0
Lung	5.5
Lymph Nodes	1.0
Nervous System	1.0
Ovary	1.5
Pancreas	1.0
Pleura	0.5
Prostate	1.0
Skin	0.5
Soft Tissue	0.5
Stomach	1.0
Testis	0.5
Thymus	0.2
Thyroid	1.0
Uterus	1.0

High-quality Datasets

From Foundational Cancer Genomic Studies

The Cancer Genome Atlas Program (TCGA)

 **NATIONAL CANCER INSTITUTE**
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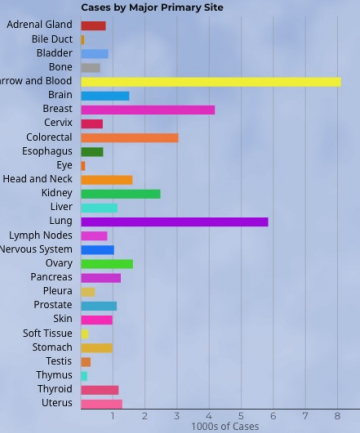

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Cases by Major Primary Site

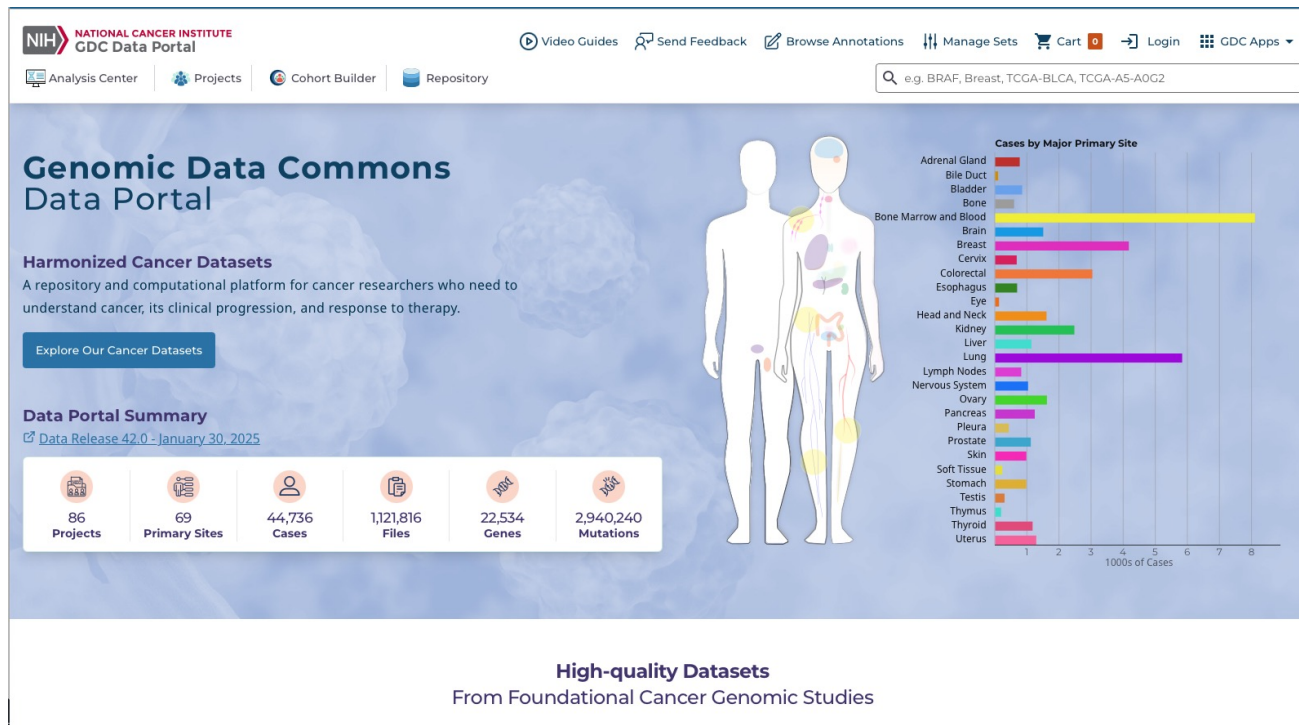


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High-quality Datasets
From Foundational Cancer Genomic Studies

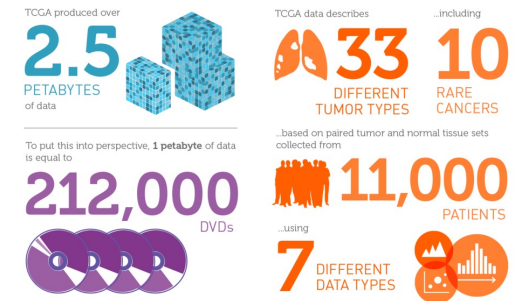
<https://portal.gdc.cancer.gov>

The Cancer Genome Atlas Program (TCGA)

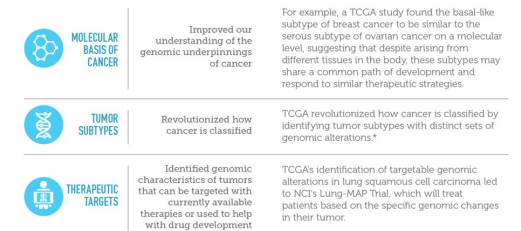


NATIONAL CANCER INSTITUTE THE CANCER GENOME ATLAS

TCGA BY THE NUMBERS



TCGA RESULTS & FINDINGS



THE TEAM



WHAT'S NEXT?



*TCGA's analysis of stomach cancer revealed that it is not a single disease, but a disease composed of four subtypes, including a new subtype characterized by infection with Epstein-Barr virus.

www.cancer.gov/ccg

<https://portal.gdc.cancer.gov>

The Cancer Genome Atlas Program (TCGA)

 **NATIONAL CANCER INSTITUTE**
GDC Data Portal

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[Analysis Center](#) [Projects](#) [Cohort Builder](#) [Repository](#)

Unsaved_Cohort

Cohort not saved

44,736 CASES

Unsaved_Cohort

No filters currently applied.

CORE TOOLS

**Projects**
View the Projects available within the GDC and select them for further exploration and analysis.

**Cohort Builder**
Build and define your custom cohorts using a variety of clinical and biospecimen features.

**Repository**
Browse and download the files associated with your cohort for more sophisticated analysis.

ANALYSIS TOOLS

**BAM Slicing Download**
25,270 Cases

**Clinical Data Analysis**
44,736 Cases

**Cohort Comparison**
44,736 Cases

**Cohort Level MAF**
17,771 Cases

**Gene Expression Clustering**
20,823 Cases

**Mutation Frequency**
18,640 Cases

**OncoMatrix**
18,640 Cases

**ProteinPaint**
16,508 Cases

**Single Cell RNA-seq**
36 Cases

**Sequence Reads**
25,270 Cases

**Set Operations**

<https://portal.gdc.cancer.gov>

cBioPortal

<https://www.cbioportal.org>

cBioPortal

 **cBioPortal**
FOR CANCER GENOMICS

[Data Sets](#) [Web API](#) [Tutorials/Webinars](#) [FAQ](#) [News](#) [Visualize Your Data](#) [About](#) [cBioPortal Installations](#) [❤️ Donate](#)

[Query](#) [Quick Search](#)

Please cite [cBioPortal](#)

Select Studies for Visualization & Analysis:

492 studies available (321114 samples)

PanCancer Studies11

Pediatric Cancer Studies14

Immunogenomic Studies8

Cell lines3

PreCancerous Studies2

Adrenal Gland3

Ampulla of Vater1

Biliary Tract16

Bladder/Urinary Tract24

Bone4

Quick select: [TCGA PanCancer Atlas Studies](#) [Curated set of non-redundant studies](#)

Looking for **AACR Project GENIE**, the largest public clinicogenomic cancer dataset? [It's available here.](#)

PanCancer Studies

☐ MSK-CHORD (MSK, Nature 2024)25040 samples

☐ MSK-IMPACT Clinical Sequencing Cohort (MSK, Nat Med 2017)10945 samples

☐ Metastatic Solid Cancers (UMich, Nature 2017)500 samples

☐ MSS Mixed Solid Tumors (Broad/Dana-Farber, Nat Genet 2018)249 samples

☐ SUMMIT - Neratinib Basket Study (Multi-Institute, Nature 2018)141 samples

☐ TMB and Immunotherapy (MSK, Nat Genet 2019)1661 samples

☐ Tumors with TRK fusions (MSK, Clin Cancer Res 2020)106 samples

☐ Cancer Therapy and Clonal Hematopoiesis (MSK, Nat Genet 2020)24146 samples

☐ China Pan-cancer (Origimed, Nature 2022)10194 samples

☐ Pan-cancer analysis of whole genomes (ICGC/TCGA, Nature 2020)2922 samples

☐ MSK MetTropism (MSK, Cell 2021)25775 samples

Pediatric Cancer Studies

<https://www.cbioportal.org>

Survival Genie 2

 Survival Genie 2.0

Analysis ID: 3d839823dcc04006adbac6ca316aec00   More ▾

I Have an Analysis ID:

Retrieve

Select your input type to get started.

Single Gene

Multiple Genes

Gene Ratio

L-R Pairs

Gene Set

Gene Set Contrast

Cluster Markers

Co-expression Network

Gene Regulatory Network

Survival outcomes are predicted by stratifying samples into high versus low groups based on the FPKM expression of the gene of interest in up to 5 datasets.

Example Output Page

Begin

Hide Tutorial

Input Overview

Analysis Specific Input

Select Dataset

Analysis Parameters

Input Review

Input Overview

Analysis ID

Each Analysis will be assigned with an analysis ID, which is shown on the top of each page.

Record the analysis ID to retrieve or share previous results (each analysis is valid for 24 hours)

Analysis-Specific Input

Single Gene

Input a gene symbol

Gene Ratio

Input two gene symbols to calculate ratio

Gene Set

Select a built-in gene-set or input a gene set by gene names

Cluster Marker

Select example data or upload a marker file through Uniprot or ICGC

Select Dataset(s)

Select up to 5 datasets for Single Gene analysis, Gene Ratio analysis, GeneSet Analysis; Select One dataset for Cluster Marker and Co-Expression Network analysis.

Other Parameters

Select a cutpoint determination method to separate high/low risk groups; Select the result type; Overall survival or Event-Free survival; Select the Signature cell type sets to calculate the correlation.

Review Inputs

Review all input parameters and submit the task

Start Analysis

Output

<https://bhasinlab.bmi.emory.edu/SurvivalGenie2/home>

GEO (Gene Expression Omnibus)

<https://www.ncbi.nlm.nih.gov/gds/>

GEO (Gene Expression Omnibus)

Gene Expression Omnibus

GEO is a public functional genomics data repository supporting MIAME-compliant data submissions. Array- and sequence-based data are accepted. Tools are provided to help users query and download experiments and curated gene expression profiles.

Getting Started

- [Overview](#)
- [FAQ](#)
- [About GEO DataSets](#)
- [About GEO Profiles](#)
- [About GEO2R Analysis](#)
- [How to Construct a Query](#)
- [How to Download Data](#)

Tools

- [Search for Studies at GEO DataSets](#)
- [Search for Gene Expression at GEO Profiles](#)
- [Search GEO Documentation](#)
- [Analyze a Study with GEO2R](#)
- [Studies with Genome Data Viewer Tracks](#)
- [Programmatic Access](#)
- [FTP Site](#)
- [ENCODE Data Listings and Tracks](#)

Browse Content

Repository Browser	
DataSets:	4348
Series:	 253004
Platforms:	27299
Samples:	7792875

Information for Submitters

Login to Submit	Submission Guidelines	MIAME Standards
	Update Guidelines	Citing and Linking to GEO
		Guidelines for Reviewers
		GEO Publications

<https://www.ncbi.nlm.nih.gov/gds/>

GTEx Portal



[About Adult GTEx](#) [Publications](#) [Access Biospecimens](#) [FAQs](#) [Contact](#)



The Genotype-Tissue Expression (GTEx) Portal is a comprehensive public resource for researchers studying tissue and cell-specific gene expression and regulation across individuals, development, and species, with data from 3 NIH projects.



The Adult GTEx project is a comprehensive resource of WGS, RNA-Seq, and QTL data from samples collected from 54 non-diseased tissue sites across ~1000 adult individuals.

[Explore >>](#)



The Developmental GTEx (dGTEx) project is a new effort to study development-specific genetic effects on gene expression and to establish a new data analysis and tissue biobank resource.

**Data Not Yet Available*

[Explore >>](#)



The Non-Human Primate Developmental GTEx (NHP-dGTEx) project is a complement to dGTEx in 2 translational non-human primate model species: the rhesus macaque and common marmoset.

**Data Not Yet Available*

[Explore >>](#)

www.gtexpportal.org

GTEx Portal

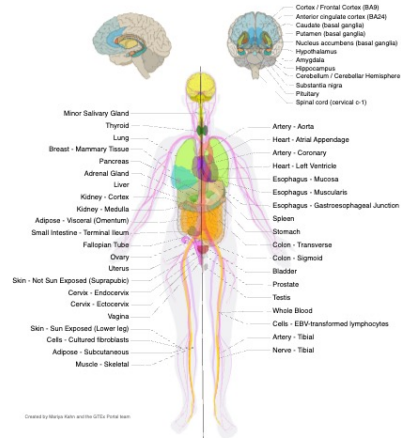
Whole Blood

Data Source: GTEx Analysis Release V10 (dbGaP Accession phs000424.v10.p2)

UBERON UBERON:0013756

Main Sampling Site

Femoral vein; subclavian vein and heart are other possible sites.



[Histology Viewer](#) | [Expression](#) | [IGV](#) | [Anatomogram](#)

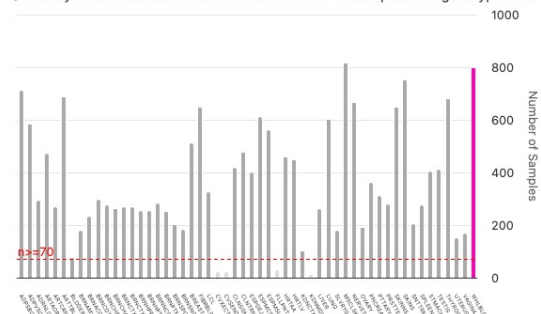
Bulk Tissue RNA-Seq Sample Info

Table: Sample count in Whole Blood

Samples	All	Female	Male
Total	803	262	541
With Genotype	800	261	539

Chart: GTEx samples by tissue

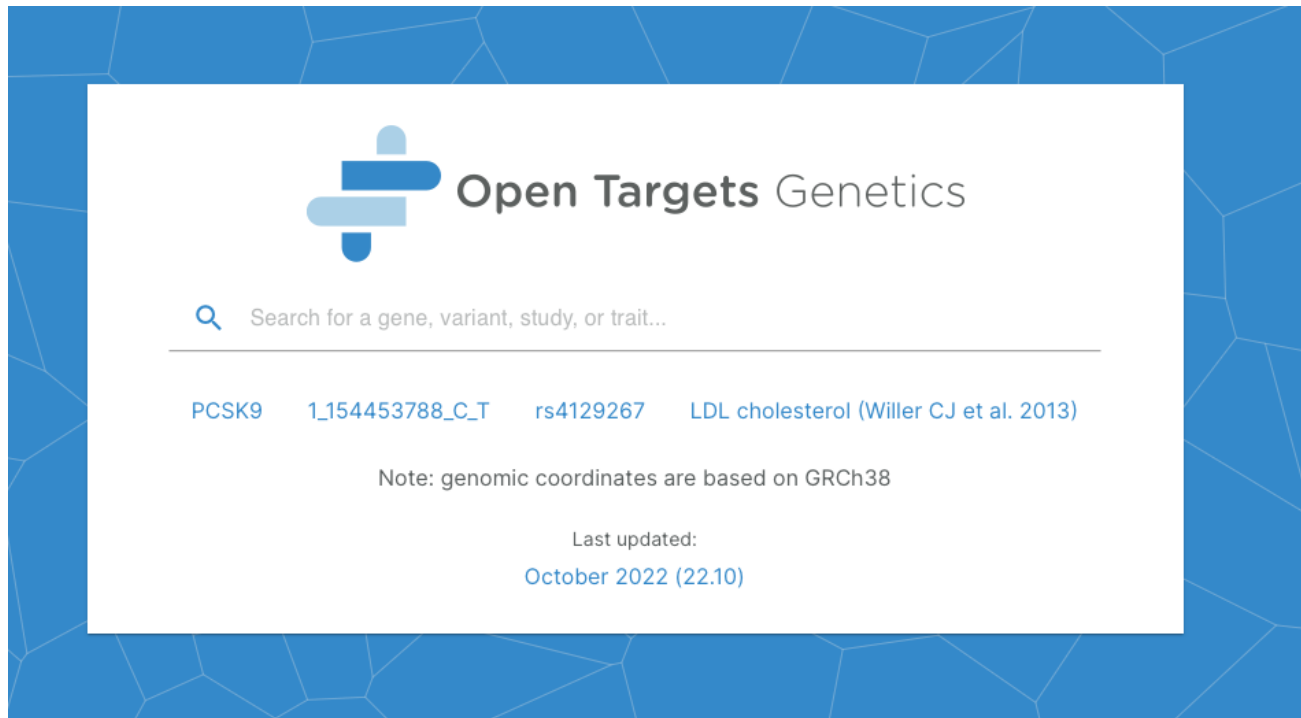
QTL analysis results are available for tissues with >70 samples with genotype data



Open Targets Genetics

<https://genetics.opentargets.org>

Open Targets Genetics

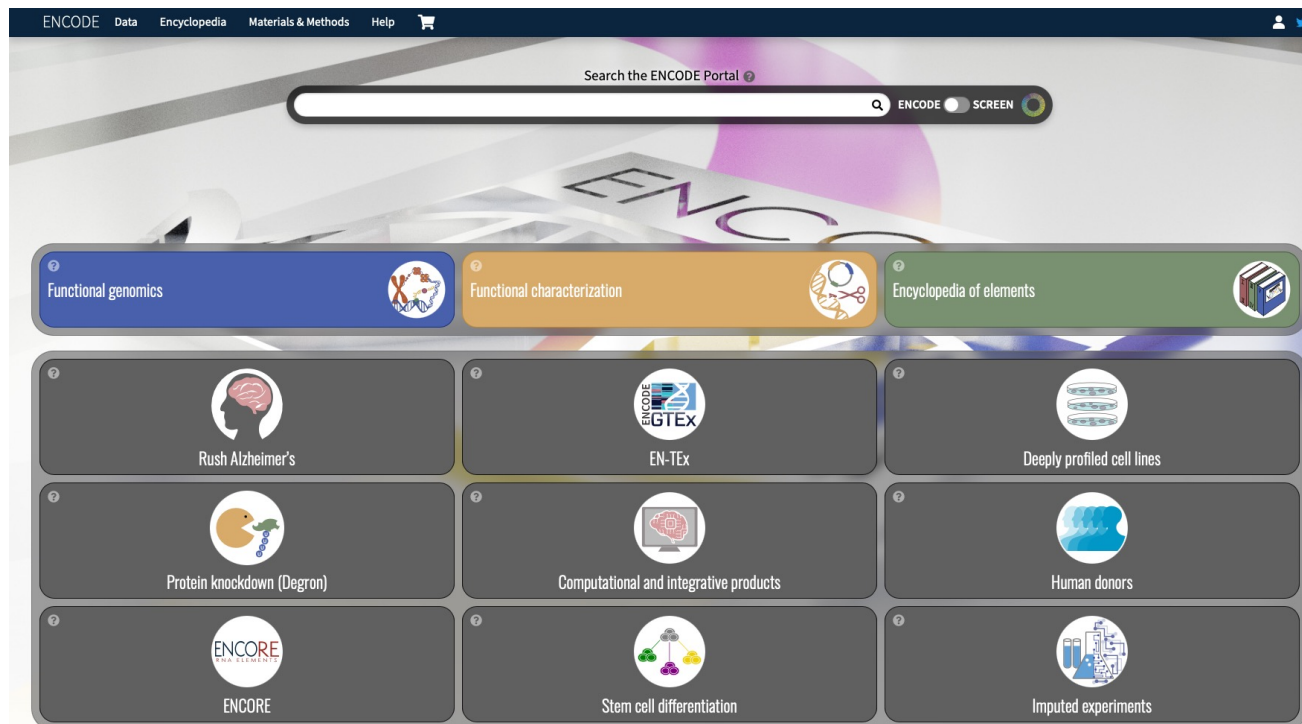


<https://genetics.opentargets.org>

EnCode

<https://www.encodeproject.org>

EnCode



<https://www.encodeproject.org>

EnCode

<https://www.encodeproject.org>

ENCODE Data Encyclopedia Materials & Methods Help

Search...

Experiment Matrix

EXPERIMENTS

Clear all selections ✕

Filter the experiments included in the matrix:

Enter search term(s)

Showing 19996 results

List Report Download Visualize

ASSAY →

← BIOSAMPLE

Assay

Assay type

Assay title

Search

TF ChIP-seq 4690
Histone ChIP-seq 3694
DNase-seq 3425
total RNA-seq 1365
mint-ChIP-seq 895
polyA plus RNA-seq 774
ATAC-seq 464
microRNA-seq 412
long read RNA-seq 397
snRNA-seq 395

Target category

Target of assay

Search

H3K4me3 786
H3K27ac 696
H3K27me3 657
H3K4me1 635
H3K36me3 628
H3K9me3 589
CTCF 516
POLR2A 187
H3K9ac 186
H3K4me2 132

	TF ChIP-seq	Histone ChIP-seq	DNase-seq	total RNA-seq	Mint-ChIP-seq	polyA plus RNA-seq	ATAC-seq	microRNA-seq	long read RNA-seq	snRNA-seq	snATAC-seq	intact Hi-C	eCLIP	DNase array	small RNA-seq	WGBS	PRO-cap	ChIA-PET	RAMPAGE	RNA microarray	genotyping array	CAGE	microRNA counts
tissue	439	2175	1605	453		393	225	305	310	390	292	110	2	121	67	162	141	1	104	2	7	17	101
dorsolateral prefrontal cortex	59	188	110	120				43	13			2											
heart	6	81	41	21		16	7	25	17	49	17	7			1	10			2				8
adrenal gland	8	39	35	27		10	8	23	26	59	15	3	2	5	2	5	3		4				2
heart left ventricle	16	84	14	9		2	15	5	4	54	65	10		3		4	6		2				2
liver	42	91	29	3		20	8	7		8	8			1	1	9	5		2				7
cell line	2831	773	583	136	54	186	123	39	57	2	32	161	250	87	110	18	27	108	29	67	73	50	8
K562	749	19	59	17		15	4	2	4		3	8	145	3	7	1	7	9	1	9	2	9	1
HepG2	814	15	3	6		11	2	2	3		3	3	105	3	3	2		4		6	2	6	1
GM12878	188	15	4	5		14	2	2	4		3	68		3	6	1		6	1	7	2	6	1
MCF-7	150	18	66	1		4	1	2	1		3	1		2	7			4		2	2	3	1
HEK293	204	6												2			1	2		1	2		
primary cell	74	506	1194	654	715	132	76	52	6	1	29	26		38	24	7	25	60	16	57	37	30	1
CD4-positive, alpha-beta T cell	1	12	160	191	6	2	1	1			1	2						2					
T-cell		11	75	51	18			7								1	7	4					
stimulated activated CD4-positive, alpha-beta T cell			133	6																			
naive thymus-derived CD4-positive, alpha-beta T cell		16	20	6	74	8	2	2			2	2						2					
CD14-positive monocyte	2	21	39	12	41	7					2	2			1	1						1	

TOPMed: Trans-Omics for Precision Medicine Program

topmed.nhlbi.nih.gov

TOPMed: Trans-Omics for Precision Medicine Program

Trans-Omics for Precision Medicine | TOPMed

What is TOPMed?

Trans-Omics for Precision Medicine (TOPMed), is a program of the National Heart, Lung and Blood Institute (NHLBI), a part of the National Institutes of Health, which aims to improve scientific understanding of the **fundamental biological processes** that underlie heart, lung, blood, and sleep (HLBS) disorders and advance precision medicine in ways that lead to disease treatments tailored to individuals' unique genes and environments.

TOPMed supports these scientific advances through the integration of whole-genome sequencing (WGS) and other omics data (e.g., metabolic profiles, epigenomics, protein and RNA expression patterns) with molecular, behavioral, imaging, environmental, and clinical data from pre-existing parent studies that have large samples of human subjects with rich phenotypic characterization and environmental exposure data. TOPMed also collects environmental and behavioral data, such as dietary habits, physical activity, and socioeconomic factors, to provide a more comprehensive understanding of the factors that contribute to these disorders.

NHLBI Artificial Intelligence Initiative (NHLBI-AI)

The NHLBI-AI Initiative was created to stimulate the development of AI methods and models for discovery research that significantly impact understanding health and the prevention, diagnosis, and treatment of heart, lung, blood, and sleep (HLBS) conditions. In addition, the Initiative will develop infrastructure to support studies involving AI by the broader research community. NHLBI-AI began in FY2024 with the establishment of an AI Coordinating Center (AICC) tasked with managing the logistics of building the consortium and bringing AI experts into collaboration with subject matter experts on existing NHLBI programs, such as TOPMed and BioData Catalyst. We envision that the NHLBI-AI Initiative will be composed of the Data Science Center (DSC), a Precision Medicine Coordinating Center (PMCC), several modular precision medicine programs, and a series of independently funded AI research projects supported by these cores. The PMCC will support the transformation, harmonization, and sharing of extant and new data so that NHLBI-AI investigators can pose a greater scope of clinically impactful research questions.

topmed.nhlbi.nih.gov

All of Us

DIVERSITY IN HEALTH RESEARCH MATTERS.

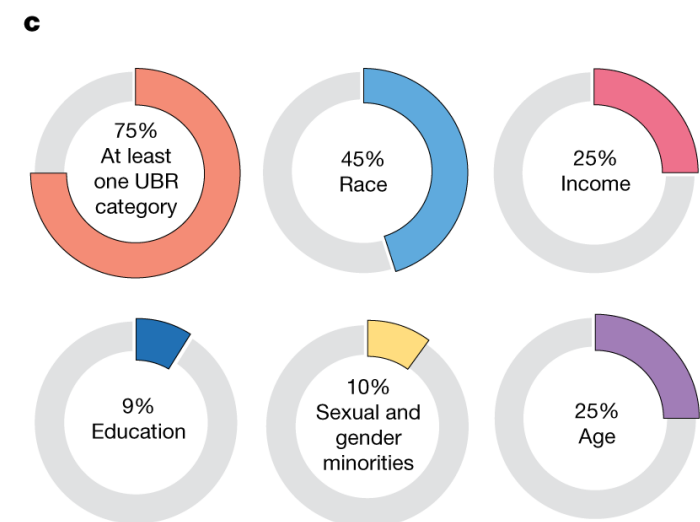
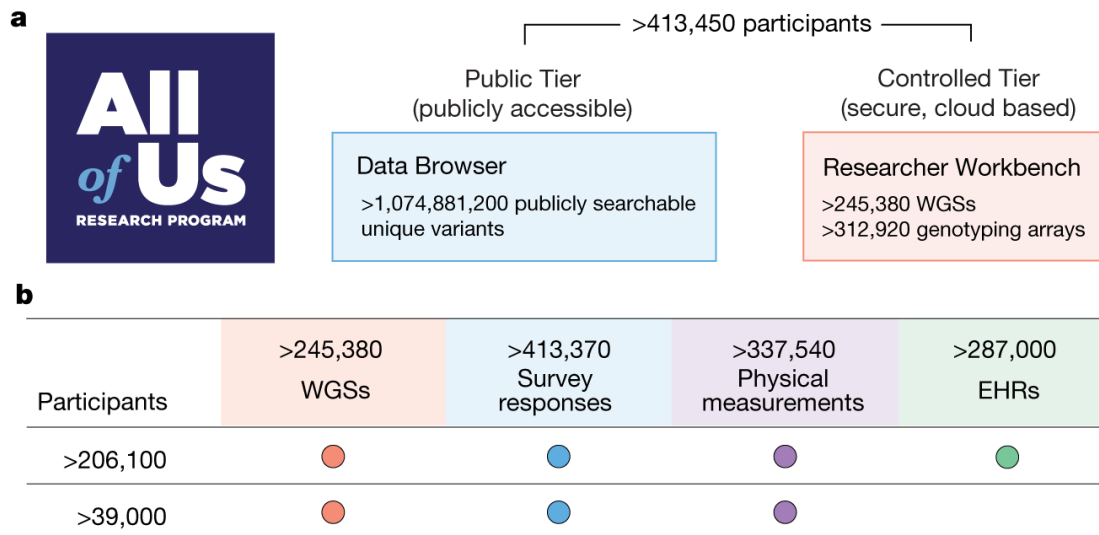
Thanks to the size and diversity of the *All of Us* dataset,
we have a better understanding of
heart health in Hispanic/Latino communities.



Reference: Montanez-Valverde, R., et al. (2022). Revisiting the Latino Epidemiologic Paradox: An analysis of data from the *All of Us* Research Program. *Journal of General Internal Medicine*. 37(15), 4013-4014.

<https://allofus.nih.gov>

All of Us



The All of Us Research Program Genomics Investigators. Genomic data in the All of Us Research Program. *Nature* **627**, 340–346 (2024)

All of Us



Surveys	• 633k+ participants with data from The Basics, including data about life functioning and granular participant-reported population descriptors • Open surveys with data available: » Overall Health » Lifestyle » Health Care Access and Utilization » Personal and Family Health History* » Social Factors of Health » Emotional Health History and Well-Being » Behavioral Health and Personality	→
Physical Measurements	• 509k+ participants with physical measurements collected in person, recorded in EHR data, and/or reported by participants, such as: » Height » Weight » BMI » Hip circumference » Waist circumference » Blood pressure » Heart rate	→
Genomics	• 447k+ participants with array data • 414k+ participants with short-read whole genome sequencing data (srWGS)** » 97k+ participants with structural variant data » All by All tables (genome- and phenome-wide analysis) • 2.7k+ participants with long-read whole genome sequencing data (lrWGS)	• 500k+ srWGS • 10k+ participants with multi-omics » RNA sequencing » Proteomics data » lrWGS
Data Linkages	• American Community Survey data linked at 3-digit zip code level	• Mortality • Environmental data • Claims • Imaging
Electronic Health Records (EHR)	• 393k+ participants with EHR data from health care provider organizations and/or provided by participants, such as: » Conditions » Procedures » Drugs	• Extracted notes

<https://www.researchallofus.org>

All of Us



Step 1

Complete Registration

RT

Registered Tier data

Once registered, you'll have access to:

 Individual (not aggregated) data

 Identifying information removed

 Electronic health records

 Survey responses

 Physical measurements

 Wearable devices

✓

Turn on Google 2-Step Verification
Completed on: Aug 12, 2025

✓

Verify your identity
Completed on: Aug 12, 2025

✓

Complete *All of Us* Responsible Conduct of Research Training
Completed on: Aug 13, 2025

Step 2

Additional Data Access

CT

Controlled Tier data - *Optional*

You are eligible to access Controlled Tier data.

In addition to Registered Tier data, the Controlled Tier curated dataset contains:

 Genomic data

 Additional demographic details

✓

The Wistar Institute must sign an institutional agreement

✓

The Wistar Institute must allow you to access Controlled Tier data

✓

Complete *All of Us* Controlled Tier training
Completed on: Sep 1, 2025

<https://www.researchallofus.org>

